

Demographics of Adult Heritage Language Speakers in the United States: Regional and Typological Differences and their Implications

Tomonori Nagano
LaGuardia Community College
The City University of New York
Education and Language Acquisition Department
31-10 Thomson Avenue (B-234FF)
Long Island City, NY 11101
Email: tnagano@lagcc.cuny.edu

ABSTRACT

Heritage language (HL) speakers have received scholarly attention in recent years as an interdisciplinary research theme, but relatively less attention has been paid to their demographics. Existing studies of HL speakers' demographics often focus on young children in areas of high immigrant concentration (i.e., California, Florida, and New York), and no study has systematically investigated cross-regional and chronological demographic patterns of adult HL speakers. From the perspective of HLs as a national resource, such demographic data on adult HL speakers are useful to gauge the availability of the bilingual workforce and determine structures needed to support and develop a bilingual U.S. population. Using the *Integrated Public User Microdata Series* (Ruggles & Sobek, 1997), which is based on data from the U.S. Census and the American Community Survey, this study makes geographical and chronological comparisons among groups of adult HL speakers from 1980 to 2010. The data show major differences in the demographics of adult HL speakers in different regions. The analyses also reveal typological differences, specifically between the adult HL speakers of Spanish and other languages. Implications of these patterns are discussed.

Keywords: heritage language; demographics

Heritage language (HL) speakers have drawn much scholarly attention in recent years as an interdisciplinary research theme among language educators, linguists, and policy makers. For language educators, understanding how to teach mixed classes with HL speakers and non-heritage second language learners in their foreign language classrooms is a perennial issue. It is a challenge because these two groups of language learners have different instructional needs and goals for learning the target language. Unless the curriculum is specifically developed for HL speakers, their unique linguistic and instructional needs are often overlooked in the language classroom. According to linguists, HL speakers contribute to establishing theoretical links between studies on first language acquisition (FLA) and second language acquisition (SLA). Research findings on HL speakers, who have undergone the acquisition processes of both first language learners and second language learners, shed light on the similarities and differences between FLA and SLA. To policymakers, HLs are national resources for the country, and their maintenance requires proactive educational programs, such as bilingual programs, and multilingual public services, such as bilingual ballots.

Over the last two decades, a considerable number of studies have been conducted on HLs from a wide range of research orientations and broad perspectives. However, in contrast with these active areas of the HL studies, relatively little is known about the demographics of HL speakers. Knowledge about the HL speakers' demographics is often based on anecdotal and experiential facts from areas with a high immigrant concentration, which tend to have a larger number of HL speakers (such as California and Florida). The dearth of studies making cross-regional comparisons and chronological comparisons of HL speakers is particularly prominent.

A purpose of this article is to provide systematic and detailed demographic descriptions of adult HL speakers in the United States using data available from the U.S. Census and the American Community Survey (ACS) from 1980 to 2010.

IMPORTANCE OF THE STUDY

A careful study about the demographics of adult HL speakers is necessary to demonstrate the significance of HL research and to understand the magnitude of benefits that the research findings bring to the general public. Such information is also crucial for policymakers and educational institutions to make decisions about bilingual services, such as creating bilingual ballots, establishing school curricula, and developing language programs and community services that reflect regional linguistic diversity. Finally, the data will be helpful to gauge the current availability of bilingual workforce, whose importance has increasingly been emphasized as the world economy has become highly integrated globally.

In this study, demographic information of adult HL speakers of nearly forty languages for each state based on census data from 1980 to 2010 are presented. This study focuses on adult HL speakers (i.e., HL speakers who are at age of 18 or older) because, from the perspective of HLs as a national resource (Brecht & Ingold, 2002), adult HL speakers' demographics directly reflect the current availability of the bilingual workforce in the United States. In addition, data on adult HL speakers are also useful to capture the chronological transitions of HL speakers because their language profiles are much less likely to change than those of young children, which are known to change to English in a short time period.

In terms of the absolute number of HL speakers, the data support the anecdotal and

experiential estimations that a large number of HL speakers reside in areas that receive immigrants, such as California, New York, New Jersey, Texas, Illinois, and Florida. However, analyses of the fine-grained distribution of HL speakers at the state/county level reveals marked regional variations and different degrees of dispersion by both language and region. Over the last thirty years, the growth rate of HL speakers has been much faster than that of the general U.S. population. However, the growth rate also exhibits regional variations, and some states have even had a decline in the number of HL speakers.

The following findings regarding the synchronic demographic patterns and diachronic changes of HL speakers over the past few decades are discussed in this study:

- (1) The number of HL speakers grew at a considerably faster rate (26.98% per decade from 1980–2010) than the average growth of the U.S. population (10.88% per decade from 1980–2010). However, the growth rates radically differ from state to state. In some states, there was even a decline in the number of adult HL speakers.
- (2) Spanish and Chinese remain the two most common groups of HLs in the United States, and their prevalence has grown rapidly over the last 30 years. In addition, although their numbers are still small in relation to the absolute number of speakers, new HLs such as Arabic, Hindi, Dravidian, Vietnamese, Russian, Amharic/Ethiopian, and Tibetan, also experienced substantial growth. At the same time, languages such as French, German, Italian, Greek, Yiddish, and Dutch are experiencing rapid declines.
- (3) Unsurprisingly, states that are typically considered immigration hubs (e.g., California, Texas, New York, Florida, and New Mexico) have large numbers of HL speakers. However, in terms of the proportion of HL speakers in the state's population, some other states (e.g., Arizona, Hawaii, Massachusetts, Nevada, and New Jersey) have proportions of HL speakers as large as those in states with high numbers of immigrants.
- (4) The majority of HL speakers are speakers of Spanish, but on the state level, there are quite a few exceptions, such as Alaska (Aleut Eskimo is the most common HL), Hawaii (Filipino/Tagalog and Japanese), Louisiana (French), Maine (French), New Hampshire (French), North Dakota (German), South Dakota (Siouan languages), and Vermont (French).
- (5) Over the last 30 years, the concentration of HL speakers has moved from metropolitan centers to suburbs or non-metropolitan areas. However, some of the HLs continue to have specific regional concentrations.

PREVIOUS STUDIES

One of the largest surveys of HL speakers to date is *the National Heritage Language Survey*, conducted by the National Heritage Language Resource Center (NHLRC) in 2007–2009 (Carreira et al., 2009; Carreira & Kagan, 2011). The NHLRC survey asked a wide range of questions, including HL speakers' language usage and proficiency, transition in the use of their primary languages between childhood and adulthood, motivation, language contact in the HL, self-assessment of their HL knowledge, self-assessment of their English knowledge, and two open-ended questions about their attitude toward the HL. The survey was administered in HL classes in postsecondary institutions, and 1,732 HL speakers responded to the survey.

Among other findings, the results of the survey highlight the critical role of country of birth and age of arrival to the United States in the maintenance of HLs. The results show that speakers with an earlier age of arrival and those born in the United States have a higher tendency to show lower self-reported proficiency in their HL. In contrast, those who arrived in the United States at the ages of 11–13 years old and 14–18 years old reported significantly higher self-reported proficiency in their HL. Age of arrival also correlates with use of the HL. Whereas those who arrived in the United States before the age of two years primarily use English to speak with friends (approximately 70–80% of the time), those who arrived between the ages of 2 and 10 years use English a little bit less frequently (60–70%); instead, they frequently employ a mix of English and the HL (approximately 30% of the time). Among those who arrived in the United States after the age of eleven, their use of the HL increases considerably and they use a mix of English and the HL more than 50% of the time. In addition, the survey revealed several intra-language differences among different HLs. For example, whereas Spanish and Russian HL speakers tend to maintain exposure to their HLs even after their arrival in the United States, HL speakers of Korean, Chinese (both Mandarin and Cantonese), Tagalog, and Vietnamese have significantly reduced opportunities to use their HLs after their arrival in the United States. For some of the languages in the latter group (e.g., Korean and Chinese), community heritage schools and religious institutions play a significant role in maintaining HL proficiency because their ethnic communities are much smaller and more scattered than the communities of the former groups. Finally, although all HL speakers rated their HL oral skills higher than their literacy skills, the two language groups that maintain regular contact with their HL (i.e., Spanish and Russian) reported a higher self-rated proficiency in their HLs than did the Korean, Vietnamese, Persian, Tagalog, and Chinese groups.

The NHLRC HL Survey provides valuable statistical data that corroborate intuitive and anecdotal knowledge about the HL speakers, but it also faces some limitations. One major problem is that, despite the high number of respondents to the survey, the majority (72%) of respondents were from one state (California) and, as a consequence, the data may not necessarily reflect the national trends of HL speakers. For example, the language groups of HL speakers appear to reflect the immigrant communities on the West Coast. The survey respondents included Armenian (3.4% of the respondents), Persian (3.5%), Tagalog (6.5%), and Vietnamese (6.6%) speakers, who may not be found as frequently in other regions. The analyses of the Census/ACS data reveal many regional differences among HL speakers and suggest that these language communities are typical HL groups in the West Coast states but not in other regions.

Rumbaut et al. (2005, 2009) reported on another set of major surveys on the HL speakers. Both *Immigration and Intergenerational Mobility in Metropolitan Los Angeles* (IIMMLA) and *Children of Immigrants Longitudinal Study* (CILS) are large-scale surveys of HL speakers in Southern California and South Florida. The IIMMLA is a cross-sectional study that sampled 5,000 first through fourth generations of immigrants in their 20s and 30s in the Los Angeles area. The survey found that the majority of the first-, 1.5-, and second-generation immigrants¹ used their HL at home (97.4%, 92.9%, and 83.5%, respectively), but there were huge declines in the HL proficiency from the first generation (86.9% speak the HL very well) to the 1.5 and second generations (46.6% and 36.1%, respectively), and then to the third generation (11.9%). Proficiency in and preference for English exhibited a reverse pattern. Only 17.7% preferred to speak English among the first generations, whereas the corresponding figures jumped to 60.7%–73.4% among the 1.5 and second generations and to 97% among the third generation.

The IIMMLA survey was supplemented by the CILS, another large-scale longitudinal survey in San Diego and Miami/Fort Lauderdale (Rumbaut, 2009). The CILS survey was administered to 5,262 speakers of HLs at the age of 14 or 15 in 1992 with two follow-up interviews in 1995 (when the respondents were 17 or 18) and in 2001–2003 (when the respondents were in their mid-20s). The results of the CILS survey clearly exhibit strong linguistic assimilation among children of immigrants: The proportion of 1.5-generation children who spoke English fluently increased from 42.4% (at 14 or 15 years old) to 54.0% (at 17 or 18 years old) and then to 72.4% (in their mid-20s). Notably, fluency in the HLs also showed improvement: 50.4% of 1.5-generation children spoke their HLs fluently at 14 or 15 years old, which increased to 49.3% (at 17 or 18 years old) and 55.8% (in their mid-20s). Rumbaut (2009) noted a clear typological difference in the HLs in terms of the degree of linguistic assimilation. The respondents who spoke Spanish had a better chance of maintaining their HL (78.3% of them understood the HL “very well” in their mid-20s), whereas a much smaller proportion of those who spoke Asian languages maintained their HLs (40.3%). Using the term “linguistic graveyard,” Rumbaut (2009) noted an extremely strong pressure for linguistic assimilation in the United States, which practically extinguishes almost all HLs within the three generations of immigration.

In addition to these large-scale surveys, several small-scale studies have focused on specific regions or languages. Lopez (1996) conducted a study of languages spoken in Greater Los Angeles using U.S. Census data in 1980 and 1990. The examination of census data suggests a daunting rate of language attrition among 1.5-generation (i.e., those who came to the United States before the age of ten) and second-generation immigrants, particularly those who speak Asian languages. According to Lopez (1996), 30% of second-generation Spanish speaking immigrants speak only English at home. In contrast, the percentages of English monolingual second-generation immigrants among Asian language speakers are noticeably high (77% for all Asian languages and 65% for Chinese speakers). The data suggest that concentration in terms of sheer numbers (i.e., the larger a linguistic community is, the slower the language attrition process becomes) and isolation (i.e., the more self-contained a linguistic community is, the slower the language attrition process becomes) are the two factors that facilitate successful maintenance of HL in Spanish-speaking communities.

Rapid language attrition, particularly among Asian school-age immigrant children, was also reported by Alba et al. (2002), who conducted multivariate analyses of the 1990 census data. Their study investigated the influence of various explanatory variables on the Anglicization process, including parental education, self-employment status, exogamous marriage, the presence of HL speakers in the household, proximity to the ethnic concentration, and the percentage of the speakers of the HL in each region. The major finding was that Asian HL speakers (e.g., Chinese and Filipino) tend to become English monolinguals faster than Hispanic immigrants (e.g., Mexican and Cuban Spanish speakers). Additionally, two explanatory variables emerged as the most critical factors for HL speakers to maintain their HLs; the first factor is parental endogamy/exogamy status, and the second factor is geographic location (in the ethnic community or not). According to their model, when these two factors are in disfavor of the maintenance of the HL (i.e., exogamy marriage in a non-ethnic community), the third-generation children will speak only English (completely lose their HL) 95% of the time. However, the influence of these two factors differs based on the HL groups. For Chinese HL speakers, the ethnic community matters much less than for Spanish speakers (Mexicans and Cubans) and, regardless of the presence or absence of the ethnic community near their dwelling, third-

generation children of Chinese HL speakers between exogamous parents almost always lose their HL. For Spanish speakers, on the contrary, the presence of the ethnic community alleviates the attrition rate of Spanish among the third-generation HL speakers born between exogamous parents at a significant rate. (68% in the ethnic community will lose their Spanish by the third generation, whereas the likelihood increases to 97% in non-ethnic communities.)

Garcia and Fishman (2002) investigated nearly a dozen HLs that are commonly spoken in New York City. The selection of languages strongly reflects the composition of local immigrant communities, such as Italian, Greek, Yiddish, Haitian Creole, and Caribbean Creole. In this volume, Pan (2002) conducted a small-scale survey on Chinese dialects in New York City and found that the majority of Chinese speakers in New York City did not speak Mandarin as their primary Chinese dialect. The most common Chinese dialect in his study was Min (36.6%), including sub-dialects such as Minnan and Minbei, followed by Cantonese (31.7%) and Wu (20.3%). Mandarin, which is typically considered the most popular Chinese dialect, accounted for a mere 10.4% of the 200 samples that Pan collected in Manhattan Chinatown and Flushing Chinatown. However, Mandarin is undoubtedly the most commonly spoken Chinese dialect in New York City. His data imply that these Mandarin speakers in Chinatown speak Mandarin as a secondary dialect or even as a second language.

In the same volume, Zentella (2002) shows the diversity of Spanish heritage speakers in New York City, particularly the Puerto Rican community. Although Spanish is considered to be a typical HL and HL speakers of Spanish are prevalent across the United States, Zentella argues that each region has a unique composition of Spanish-speaking communities and questions monotonous umbrella terms such as ‘Latinos’ and ‘Hispanics.’ According to her, nearly a half of the Spanish speakers in NYC are Puerto Rican (50.3%), and there are large numbers of Dominicans (18.7%), Colombians (4.7%), and Ecuadorians (4.4%). Mexicans, who typically reside in the Southwest and California, account for only 3.3% of the total Spanish-speaking community in NYC. In NYC, Caribbean dialects of Spanish, from such places as Puerto Rico, Cuba, and the Dominican Republic, are often sociolinguistically regarded as far less prestigious than the inland dialects of Spanish, from such places as Chile, Argentina, Columbia, and Mexico. For example, 60% of Puerto Rican immigrants and 80% of Dominican immigrants responded negatively to the question, “Should the Spanish of your group be the one taught in NYC schools?” This lower interest in maintenance of Spanish is one of the reasons for the faster rate of Anglicization among the Caribbean Spanish speakers. According to data about Puerto Rican families in East Harlem (Zentella, 1997), a significant number of Puerto Ricans had become English monolingual or English speakers with weak Spanish skills between 1979 and 1990 (23% English monolingual in 1979 to 71% in 1990), and 17% of Puerto Ricans reported themselves to be purely English monolingual in 1990. Although no quantitative data was presented, the rate of language attrition among speakers of more prestigious dialects of Spanish appears to be much slower than that of the Puerto Rican community.

Finally, chronological comparisons of the HL speakers’ demographics show new trends in the commonly spoken HLs in the United States. Using the census data and the survey on foreign language programs in secondary schools conducted by the Center for Applied Linguistics, Fee et al. (2014) reported recent transition in the demographics of HL speakers in the United States. The comparison between the census data in 2009 and 2011 shows a consistent pattern of Spanish as the dominant HL spoken at home in the United States (Nearly 70% of the HL speakers use Spanish at home.). However, the composition of the rest of the HLs exhibited

rather quick changes within a span of just a few years. For example, the languages of the current immigrant populations, particularly Hindi, Vietnamese, and Arabic, outnumbered the languages of the former immigration groups (French, German, and Italian, respectively) in 2011. The growth of Arabic speakers is particularly notable considering the scarcity of Arabic classes in K–12 schools. Fee et al. (2014) argue the need for a wider range of bilingual and foreign language programs in K–12 schools, which are limited to only a selected number of languages, specifically Spanish, French, German, Japanese, Chinese, and Russian.

Changes in the HL speakers' demographics have also been reported by Potowski (2010) and her colleagues. Potowski (2010) presented the demographics, history, and available media (e.g., newspapers and TV programs in the HL) of the twelve most common HLs and Native American languages in the United States. Similar to the findings of Fee et al. (2014), Potowski (2010) also noted gaps between the number of students in foreign language classes at post-secondary institutions and the number of HL speakers in the United States. However, the influence of the HLs spoken on the college-level language courses was also observed in the growth rates of enrollment. The languages that had the greatest growth in college-level language course enrollment from 2002–2006 were those spoken by recent immigrant families (i.e., Arabic (127% growth), Chinese (51%), Tagalog (37%), and Korean (37%)). Potowski (2010, pp. 17–18) proposes that a better understanding of HL populations will benefit the U.S. society through “the appreciation and promotion of linguistic diversity in the USA.”

To summarize, the earlier studies on HL speakers' demographics suggest a rapid Anglicization process that eradicates the functional proficiency of the HL by the third generation and a great degree of diversity in terms of both regions and languages. Additionally, the majority of the previous studies focused on young children (Fee et al., 2014; Alba et al., 2002). The current study will focus on adult HL speakers unlike the earlier studies that investigated the overall demographics of people who speak non-English languages at home (Shin, 2003, 2010; Ryan, 2013), young children who speak HLs in the context of their Anglicization process (Alba, 2002), and the unmet needs of bilingual programs at K–12 institutions (Fee et al., 2014).

DEFINITION OF HERITAGE LANGUAGE SPEAKERS

A clear definition of the target population is crucial to obtain their demographic information. Obviously, the demographic data on HL speakers, such as an estimated number of HL speakers, depends on how we define HL speakers, since the specificity of the definition negatively correlates with the estimated size of the target population. If HL speakers are defined with a set of highly specific conditions, such as place of birth, language input during youth, languages spoken by parents, level of HL proficiency, awareness of the HL culture, and self-identity, the estimated number of HL speakers will dwindle to a very small number. Conversely, the estimate becomes extremely large if HL speakers are broadly defined simply as people who had at least some exposure to a non-English language in the local community or at home. As several researchers have argued (Wiley, 2014), no one-size-fit-all definition for HL speakers exists, and definitions should emerge based on the purpose of and the resources available to a particular study. In this section, I review several definitions for HL speakers that are commonly adopted in the existing literature and propose one working definition for the current study.

Definitions of Heritage Language Speakers

There are essentially three different approaches to defining HL speakers. Fishman (2001) is one of the oft-cited definitions, which classifies HL speakers into three sub-groups based on the socio-historical perspectives of the United States. In his definitions, HL speakers can be classified into three sub-groups: *indigenous HL speakers* (e.g., speakers of Native American languages), *colonial HL speakers* (speakers of, e.g., French, Italian, or German), and *immigrant HL speakers* (speakers of, e.g., Spanish, Arabic, Chinese, or Japanese). Fishman's definition of HL speakers encompasses a historical account of HL speakers in the United States and is rather broadly defined. His definition presumes that each individual in the United States who speaks a language other than English can identify his/her historical roots in a community of speakers of the language.

Valdés (2001) offers another definition of HL speakers, focusing on linguistic and genealogical attributes of the HL speakers in school programs. According to Valdés, an HL speaker is “a language student who is raised in a home where a non-English language is spoken” and “speaks or at least understands the language (. . .) and is to some degree bilingual in that language and in English” (Valdes, 2001, p. 38). In this definition, HL speakers should have at least one parent who speaks a non-English language as a native language and should have obtained some fluency in the HL when they were raised and educated in an English-speaking environment. Although Valdés emphasizes that an HL speaker could merely have a minimal passive understanding in the HL because they have received primarily language input in English, her definition clearly focuses on the linguistic aspect of HL speakers. An HL speaker speaks both the HL and English; the latter is their primary language because HL speakers' dominant languages often switches from the HL to English during the pre-puberty years due to English-medium instruction at K–12 school.

Finally, Hornberger and Wang (2008) attempt to define HL speakers from psychological, sociopolitical, and socioeconomic perspectives. According to Hornberger and Wang's definition, HL speakers are “individuals with familial or ancestral ties to a language other than English who exert their agency in determining if they are HL speakers of that language” (p.6). This definition attempts to capture the identity issues of HL speakers and departs from the definitions solely based on linguistic features, such as HL proficiency. For example, some HL speakers, particularly those whose HL entails political connotations (e.g., Arabic post 9/11) or for whom the use of the HL places the speakers into a social minority group, may have different perceptions from other HL groups. Hornberger and Wang's definition weighs more on the psychological aspect of the HL, such as the ideology and self-identity between the mainstream language (i.e., English) and the HL.

It is helpful to consider these definitions of HL speakers/learners on a spectrum of two opposing extremes (Polinsky & Kagan, 2007; Carreira & Kagan, 2011). On the one side of the spectrum is the *broad* definition, for which the term *heritage language learner* is often used. On the one side of the spectrum is the *broad definition*, which emphasizes the cultural and ethnic links to their HLs and is akin to the definitions discussed above (Fishman, 2001; Hornberger & Wang, 2008). On the other side of the spectrum is the *narrow definition*, which focus on individuals' linguistic proficiency for the purpose of drawing implications for education and instruction (Valdes, 2001). Since the linguistic advantages of HL speakers/learners are often lost within the third generation of immigrants, this narrow definition often considers only the 1.5-generation, second-generation, and a handful of third-generation immigrant HL speakers.

HL Speakers and Learners of a Language Other Than English

It is also possible to define HL speakers as one of the subgroups of learners of languages other than English (LOTEs). HL speakers, particularly those who have not attained a near-native level proficiency in the HL, often mix with non-HL LOTE students who have never been exposed to the target language until it is taught in the classroom. According to a national survey (Furman, Goldberg, & Lusin, 2010), the most commonly taught LOTEs at colleges and universities in the United States are Spanish (864,986 students), French (216,419), German (96,349), ASL (91,763), Italian (80,752), Japanese (73,434), Chinese (60,976), Arabic (35,083), Latin (32,606), Russian (26,883), Greek (20,695), Hebrew (13,807), Portuguese (11,371), and Korean (8,511). These common LOTEs significantly overlap with what we typically think of as HLs. It is possible that students who are reported here as LOTE learners are actually HL speakers taking a college-level modern language course. It is not unusual to find some HL speakers in an intermediate or advanced modern language course who wish to reconnect with their ethnic heritage roots or plan to satisfy their foreign language requirement using their linguistic heritage.

One major difference between LOTE learners and HL speakers is their language acquisition process. HL speakers receive a considerable amount of input in the HL that their parent speaks at home and/or in the community. On the contrary, LOTE learners who are not HL speakers usually start learning a language at K–12 schools or post-secondary institutions, and they receive a highly structured but extremely limited amount of input in the target language. Due to these different paths to the acquisition, linguistic profiles of LOTE learners and HL speakers are different such that HL speakers tend to do well in the spoken modes (i.e., listening and speaking), whereas LOTE learners exhibit a higher proficiency in the written modes of language (i.e., reading and writing).

Several studies indicate that HL speakers are equipped with qualitatively different sets of linguistic knowledge from that of LOTE learners. For example, the advantage of HL speakers' phonological skills is attested in experimental studies (Kagan & Dillon, 2003; Montrul, 2002, 2009; Polinsky & Kagan, 2007; Polinsky, 2011). Au et al. (2002) and Oh et al. (2003) compared HL speakers and LOTE learners in the same level of language courses and found that HL speakers performed better than LOTE learners on the accent ratings and the production of stop consonants in Spanish (i.e., [p, t, k, b, d, g]) (Au et al., 2002) and in Korean (i.e., [p^h, t^h, k^h, p, t, k]) (Oh et al., 2003). However, these two groups of language learners exhibit no significant difference in their morphosyntax (grammar) tasks.

METHODS

Definition of HL Speakers for the Current Study

As previously discussed, in the research literature, HL speakers have been defined in terms of their ancestral, linguistic, or psychological identity. HL speakers exhibit a wide range of characteristics and share some similarities with and differences from LOTE learners. In sum, the following factors play key roles in defining HL speakers.

- (1) Ethnic background: According to Fishman's definition of HL (Fishman, 2001), HL speakers can be classified based on the ethnic and ancestral connection to the language.

- (2) Language proficiency: This is likely the most prototypical definition of HL speakers. Language proficiency and parents' native language are prominently expressed in Valdés's definition of HL speakers (Valdés, 2001).
- (3) Identity & sociopolitical circumstances: HL speakers can also be defined by their psychological and sociopolitical circumstances. This definition focuses on HL speakers' self-identity rather than their linguistic proficiency (Hornberger & Wang, 2008).
- (4) Age of arrival / country of birth / parents' native language: Demographically speaking, the majority of HL speakers in the United States have parents who speak a language other than English. If they are not born in the United States, HL speakers typically arrived in the United States in their youth, before puberty (Carreira & Kagan, 2011).

In establishing a definition of the HL speakers for the current study, the *availability of data* turns out to be an extremely important element to consider. The definition of HL speakers must be as precise as possible, but at the same time, the degree of specificity is confined by the available data and the purpose of making such definitions. Unless surveys are administered specifically for HL speakers as a target population (e.g., Carreira & Kagan, 2011; Mori & Calder, 2013), demographic studies using pre-existing data are limited in the options to identify HL speakers. In this study, specificity is limited to the data source, that is, questions concerning race, ethnicity, immigration age, and home language in the U.S. Census and ACS. In the previous studies using the U.S. Census data, the following variables are employed to identify HL speakers (all descriptions here are adopted from the Integrated Public Use Microdata Series (IPUMS) database (Ruggles & Sobek, 1997)):

- (1) Race and Hispanic origin: Race responses have several major selections (e.g., White, Black, American Indian or Alaska Native) and a selection of "Some other race" where the respondent can identify their own race. Hispanic origin identifies persons of Hispanic/Spanish/Latino origin and classifies them according to their country of origin when possible.
- (2) Household: Household serial number is an identifying number unique to each household record in a given sample.
- (3) Birthplace: Birthplace indicates the U.S. state, the outlying U.S. area or territory, or the foreign country where the person was born
- (4) Age: Age reports the person's age in years as of the last birthday.
- (5) Language at home: Language at home reports the language that the respondent speaks at home.
- (6) English speaking: English speaking indicates whether [and how well] the respondent is able to speak English.
- (7) Year of immigration: Year of immigration reports the year in which a foreign-born person entered the United States.

For example, Alba et al. (2002) used nativity information in the census data, such as household, birthplace, and age, to identify the first generation (foreign-born), the second

generation (U.S.-born to at least one foreign-born parent), and the third and later generations (U.S.-born to U.S.-born parents). Using the same data source, Fee et al. (2014) identified HL speakers by means of the home language variable. Rumbaut (2005) used the age of immigration variable in the IIMMLA survey data to classify HL speakers. In his study, the first-generation immigrants are defined as those who arrived in the United States at age 13 years or later, the 1.5 generation as those who arrived in the United States before age 12 years, the second generation as those who were born in the United States to two foreign-born parents, and the 2.5 generation as those who were born in the United States to one foreign-born parent.

For the most part of this study, I use the information about age, home language, English speaking ability, and year of immigration to identify HL speakers in the data. To be more specific, I selected individuals from the census data who (a) were age 18 or older, (b) spoke a language other than English at home, (c) were bilingual (i.e., did not select “Do not speak English” in the question regarding English proficiency), and (d) immigrated at an age no older than 18 years old if they were foreign born.

During the analyses, it became obvious that geographic variables have a significant influence on the demographics of adult HL speakers. To present data fine-tuned for different geographic areas, I also employed the following geographic variables in this study.

- (1) State: States are identified according to the Federal Information Processing Standards (FIPS) coding scheme.
- (2) County / Consistent Public User Microdata Area (ConsPUMA): ConsPUMA identifies the most detailed geographic areas that can consistently be identified across samples from 1980 onward. There are approximately 2,000 *Public Use Microdata Areas (PUMAs)*, which are geographic areas representing at least 100,000 people not crossing the state boundaries.

Measure

For the most part, the data are presented in simple and straightforward measures such as raw numbers (frequency), percentages, and averages. However, the Gini index, an index that I adopted as a measure for diversity, needs some explanation. The Gini index, also known as the Gini coefficient, is one of the measures of statistical dispersion similar to standard deviation, range, and quantiles. It is typically used to represent income disparity, but it can also be employed to represent any distributional dispersion. The theoretical range of the Gini index is from 0, which indicates equal distribution across all samples (no inequity), to 1, which indicates the maximum inequity, meaning all value is centered in one sample point. In this article, I use the Gini index to represent how the HL speakers are distributed in each state. (ConsPUMA are the sample unit.)

TABLE 1
Hypothetical Distribution of Heritage Languages in Four Counties

	Language 1	Language 2	Language 3	Language 4	Total
County A	40	50	200	2000	2,290

County B	40	80	100	1,000	1,220
County C	40	30	20	200	290
County D	40	20	10	100	170
Total	160	180	330	3,300	3,970
Average	40	45.00	82.50	825.00	–
Standard Deviation	0	26.46	88.08	880.81	–
Gini index	0	0.28	0.49	0.49	–

Table 1 shows a hypothetical state that consists of four counties (County A–County D) and four heritage languages (Language 1–Language 4). As seen in the Total row, each language has a different number of speakers. Language 4 has the largest number of speakers, whereas Language 1 has the smallest number of speakers. The sizes of groups of language speakers are directly reflected in the average (the total number of speakers / the number of counties). For example, in an equal distribution, there are 40 speakers of Language 1 in each county. Because the actual distribution of Language 1 is 40 speakers in each county, the standard deviation and the Gini index, the two measures of dispersion, are zero (no inequity). Language 2 has an unequal distribution of speakers, and as a result, its standard deviation and Gini index show a degree of dispersion (i.e., 26.46 and 0.28). Standard deviation is often used as the measure of dispersion, but it is difficult to make direct comparisons when the sizes of populations are different. For example, Language 3 has a significantly higher standard deviation than Language 2 (i.e., 26.46 vs. 88.08) due to its more skewed distribution of speakers. However, the high standard deviation can also be attributed to the larger quantity of speakers of Language 3 (i.e., 180 vs. 330). For this reason, the standard deviation by itself cannot be used to compare the differences in degree of dispersion between the two groups. The Gini index, however, controls for the size of the population and can be used to compare two groups of different sizes (i.e., 0.28 and 0.49). The comparison between Language 3 and Language 4 demonstrates that as long as the dispersion remains the same, the Gini index shows the same value regardless of the size of the population.

DATA AND DISCUSSION

The Data Sources

U.S. Census and ACS data from 1980 to 2010 were collected from the *Integrated Public Use Microdata Series* (IPUMS) database (Ruggles & Sobek, 1997). The U.S. Census is a decennial census that is administered among citizens, non-citizen legal residents, non-citizen long-term visitors and illegal immigrants dwelling in the United States. The questions cover a wide range of topics, including gender, age, race, ethnicity, family relationships, home ownership, education, housing, and jobs. Starting in 2005, the ACS (Gaquin & White–Dunn, 2014), an annual sample-based national survey, has been used to supplement information on education, housing and jobs and to make statistical estimates about the questions addressed in the decennial census.

Data: Languages and Races

The primary variable to identify HL speakers in this study is their home languages. Table 2 shows the list of languages reported in this study. It is worth noting that the linguistic typology adopted by the U.S. Census Bureau is sometimes criticized for its lack of linguistic accuracy. For example, all dialects of the Chinese language (e.g., Mandarin, Cantonese, Min, Wu etc.) are consolidated into one umbrella term, Chinese. Dialectal variations in Arabic and Russian are also not considered in the home language item of the U.S. Census and ACS.

TABLE 2
Language Classification in the IPUMS (U.S. Census and ACS)

Albanian	Hawaiian	Serbo–Croatian, Yugoslavian, or Slavonian
Aleut, Eskimo	Hebrew or Israeli	Slovak
Amharic, Ethiopian, etc.	Hindi and related	Spanish
Arabic	Indonesian	Sub-Saharan Africa
Armenian	Iroquoian	Swedish
Athapaskan	Italian	Thai, Siamese, or Lao
Burmese, Lisu, or Lolo	Japanese	Tibetan
Celtic	Keres	Turkish
Chinese	Korean	Ukrainian, Ruthenian, or Little Russian
Czech	Lithuanian	Vietnamese
Danish	Magyar or Hungarian	Yiddish, Jewish
Dravidian	Micronesian or Polynesian	Zuni
Dutch	Native	African Language
English	Navajo/Navaho	American Indian Language
Filipino or Tagalog	Norwegian	Near East Arabic dialect
Finnish	Persian, Iranian, or Farsi	Siouan languages
French	Polish	Other Balto-Slavic
German	Portuguese	Other East/Southeast Asian
Greek	Rumanian	Other Malayan
Hamitic	Russian	Other or not reported
		Other Persian dialects

Although there is no inherent link between race and HL, the information obtained from the race and ethnicity questions in the census is nonetheless useful to grasp the general trends of

HL speakers in the United States. There are two separate questions concerning the respondents' ethnicity in the U.S. Census. Race asks for respondents' self-reported ethnicity among the five categories: 'American Indian or Alaska Native,' 'Asian,' 'Black or African American,' 'Native Hawaiian or Other Pacific Islander,' and 'White.' Hispanic Origin asks whether the respondent is 'Hispanic or Latino Origin' or 'Not Hispanic.' Those who select 'Hispanic or Latino Origin' may identify more specific Hispanic origins.

Table 3 shows the summaries of Race and Hispanic Origin questions from the U.S. Census in 1980, 1990, 2000, and 2010. The overall increase in the population between 1980 and 2010 is 10.91%, and the only race whose population increased below the average was White (8.5%). The one single ethnic group that had by far the largest increase between 1980 and 2010 was Asian and/or Pacific Islander (63.27%). In terms of Hispanic origin, the total non-Hispanic population shows a lower growth than average (6.85%), suggesting that the growth of the U.S. population during 1980 and 2010 was largely driven by the Hispanic population. The ancestry data among the Hispanic population support the rapid growth of the Hispanic groups, showing two- to three-fold growths in all sub-groups. In sum, these figures corroborate the recent anecdotal tendencies of increasing HL courses in Spanish and Asian languages, most notably Mandarin and other Chinese dialects, in the last decade.

TABLE 3

Race/Ethnicity Data from the U.S. Census Data in 1980, 1990, 2000, and 2010

	1980	1990	2000	2010	Average Increase
Total Population	226,862,400	248,107,628	281,421,906	309,349,689	10.91%
White	194,643,640	208,834,225	231,025,096	248,564,897	8.50%
Black	26,679,320	29,821,972	35,950,767	41,380,302	15.81%
American Indian/Alaska Native	1,539,680	1,999,300	2,635,747	2,767,721	22.23%
Asian and/or Pacific Islander	3,779,800	7,211,476	11,366,992	16,074,005	63.27%
Other race, non-Hispanic	219,960	240,655	443,304	562,764	40.19%

	1980	1990	2000	2010	Average Increase
Total Population	226,862,400	248,107,628	281,421,906	309,349,689	10.91%
Not Hispanic	212,087,320	226,270,777	246,217,426	258,620,119	6.85%
Mexican	8,771,800	13,374,999	20,867,722	32,915,983	55.41%
Puerto Rican	2,035,020	2,632,326	3,400,527	4,682,531	32.08%
Cuban	822,120	1,058,497	1,248,064	1,883,599	32.53%
Other	3,146,140	4,771,029	9,688,167	11,247,457	56.93%

Data from Home Language Questions

Using the two questions concerning the home language in the U.S. Census and ACS, it is possible to make two different estimates of HL speakers in the United States. Figures 1 and 2 show the estimated numbers of HL speakers in the ten states with the highest numbers of HL speakers in 2010. The estimate in Figure 1 uses only the census question about the use of languages other than English at home, and according to this estimate, there are 59.5 million people are classified as HL speakers. Among these 59.5 million possible HL speakers, approximately 15.2 million speakers, or about one quarter of the total number of HL speakers, reside in California, 8.1 million speakers (13.6%) in Texas, and 5.47 million speakers (9.2%) in New York.

According to the definition of HL speakers that I adopt for this study, the estimate of the home language question alone is undoubtedly too broad an estimate because it may include young children and first-generation HL speakers (recent immigrants) who speak only the HL without functional English skills. Because this study is primarily interested in bilingual adult HL speakers in the United States, it is necessary to consider other factors to separate these non-target HL speakers from the target population of the current study. By combining the information on home language, age, immigration year, and self-reported proficiency in English, another estimate of HL speakers can be obtained (Figure 2). This definition follows a narrower definition of HL speakers such that the HL speakers are “adults who speak English as their primary language with some exposure to non-English languages at home.” Based on this estimate, there are 28.7 million possible HL speakers in the United States. The proportion of HL speakers in each state remains roughly the same: Slightly less than a quarter of HL speakers, or 6.2 million, reside in California, 4.15 million speakers (14.5%) in Texas, and 2.56 million speakers (8.9%) in New York.

FIGURE 1: The Estimated Number of HL Speakers for Each State and the Proportion of HL Speakers to the Total Population of All States

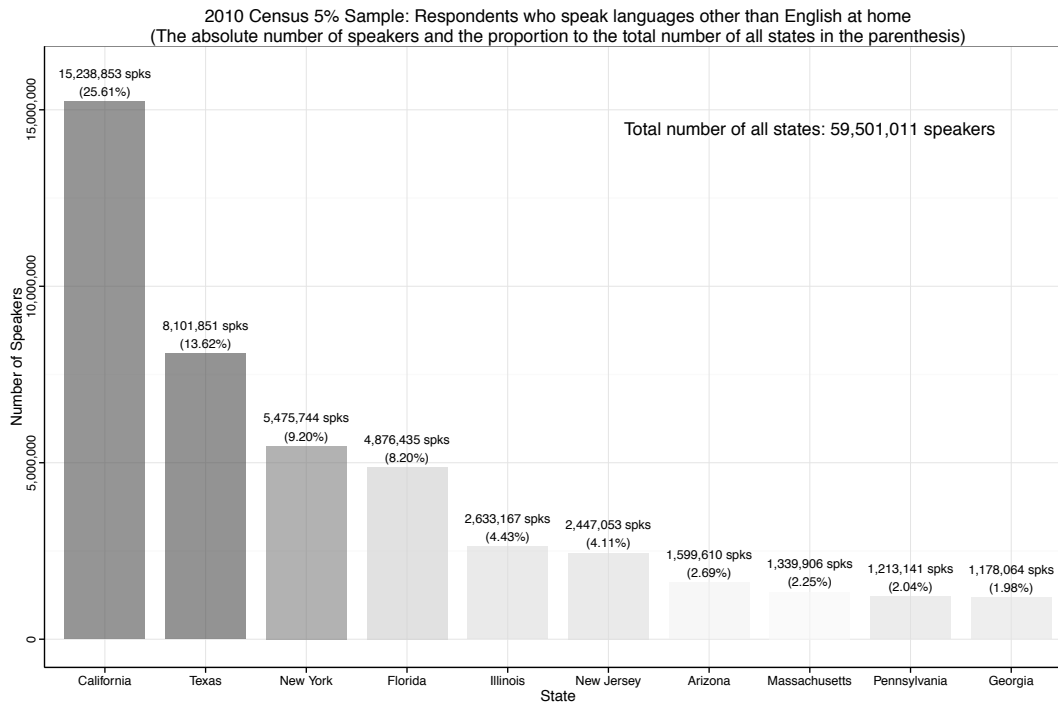


FIGURE 2: The Estimated Number of HL Speakers with Some English Proficiency for Each State and the Proportion of HL Speakers to the Total Population of All States

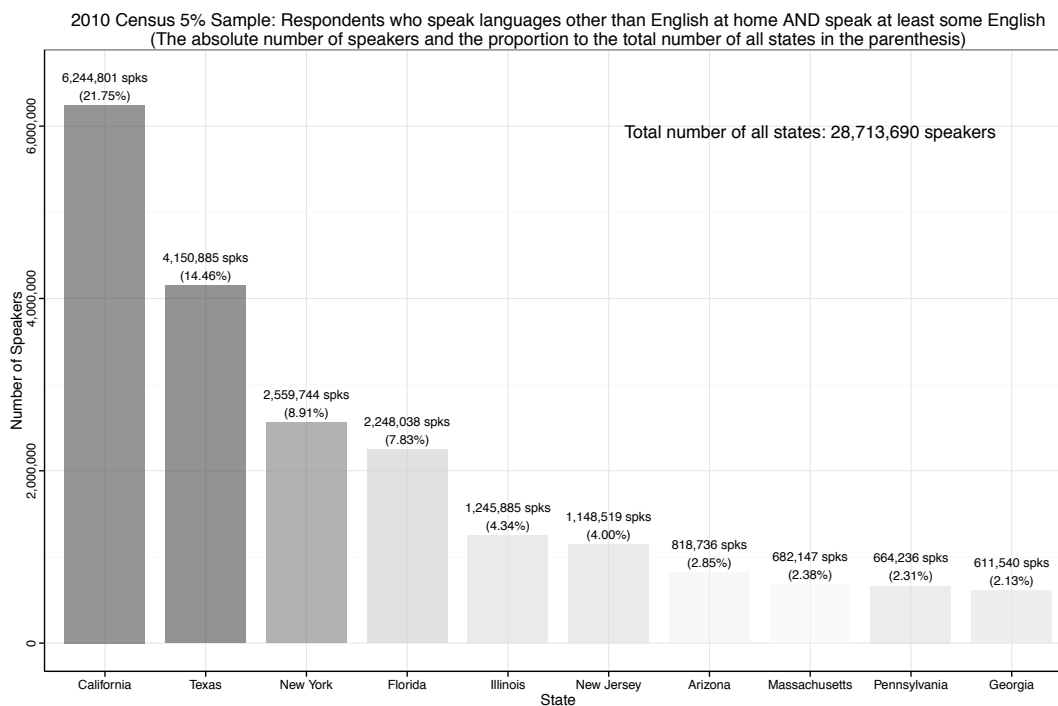


Table 4 shows the numbers of HL speakers in each state at four time periods a decade apart. The average increase of HL speakers in the U.S. population from 1980 to 2010 is 26.98% per decade, which is significantly higher than the growth rate of the total U.S. population per decade (10.88%, according to the U.S. Census). However, the growth of HL speakers is not consistent across all U.S. states. The states with large numbers of HL speakers tend to have higher growth rates. For example, California saw a slightly higher increase rate (32.05%) than the national average. The same pattern was observed in Texas (36.17%), Florida (47.63%), Arizona (40.07%), and Georgia (94.87%). Most of these states are typically considered the hubs of immigrants/ethnic communities, so it is not surprising to see such rapid increases. However, even among the states with large numbers of HL speakers, some states experienced sluggish growths, such as New York (10.90%), Illinois (20.43%), New Jersey (22.04%), Massachusetts (15.91%), and Pennsylvania (11.06%). The patterns of growth are highly inconsistent after these top ten states. Some states, such as Hawaii (0.31%), Louisiana (−6.57%), Maine (−9.8%), Montana (0.92%), New Hampshire (−4.56%), North Dakota (−23.35%), Rhode Island (−2.22%), South Dakota (−4.15%), Vermont (0.75%), and West Virginia (−0.45%), saw no increase or a decline in the number of HL speakers.

TABLE 4

The Numbers of HL Speakers of each U.S. State in 1980, 1990, 2000, and 2010

State	1980	1990	2000	2010	Average Increase Rate per Decade
California	2,869,920	4,865,328	5,605,047	6,244,801	32.05%
Texas	1,652,160	2,406,404	3,278,754	4,150,885	36.17%
New York	1,893,260	2,021,866	2,482,271	2,559,744	10.90%
Florida	700,400	1,055,856	1,613,390	2,248,038	47.63%
Illinois	718,040	844,290	1,110,040	1,245,885	20.43%
New Jersey	634,320	779,636	1,007,534	1,148,519	22.04%
Arizona	302,580	436,956	675,390	818,736	40.07%
Massachusetts	439,240	494,780	608,088	682,147	15.91%
Pennsylvania	486,920	505,422	562,470	664,236	11.06%
Georgia	89,900	174,692	437,755	611,540	94.87%
Virginia	152,320	286,095	425,421	589,677	58.38%
Washington	168,960	251,400	419,396	560,799	49.78%
North Carolina	90,800	156,825	358,547	515,301	81.69%
New Mexico	300,780	360,675	418,283	469,128	16.01%
Maryland	155,200	253,372	359,333	467,767	45.08%
Colorado	204,180	227,745	359,816	433,319	29.99%
Michigan	355,460	364,382	454,591	432,989	7.50%
Ohio	317,020	341,840	395,323	411,065	9.15%
Connecticut	250,500	257,012	305,365	362,613	13.39%
Nevada	49,240	91,559	217,238	316,210	89.59%
Minnesota	155,240	147,154	221,911	281,471	24.14%
Indiana	132,160	155,184	227,702	273,218	28.05%

Louisiana	312,920	306,345	282,450	254,745	-6.57%
Oregon	86,640	119,664	211,874	253,828	44.99%
Wisconsin	169,920	159,216	210,802	234,977	12.52%
Tennessee	57,120	84,495	163,105	220,546	58.73%
Missouri	94,820	111,350	163,144	202,600	29.38%
Utah	62,020	81,753	153,186	195,397	48.92%
Oklahoma	82,820	106,675	141,510	178,677	29.24%
Hawaii	163,560	170,786	171,955	164,768	0.31%
South Carolina	47,980	76,335	120,668	159,432	49.77%
Kansas	75,620	88,632	124,194	150,161	26.08%
Alabama	46,240	75,248	105,732	132,426	42.83%
Iowa	62,660	64,827	95,770	113,609	23.27%
Kentucky	39,060	48,258	94,962	109,302	45.14%
Arkansas	26,560	37,620	70,814	96,167	55.23%
Rhode Island	99,900	96,928	99,247	93,218	-2.22%
Nebraska	50,420	48,762	75,293	81,144	19.63%
Idaho	31,040	34,360	61,429	77,843	38.73%
Alaska	31,680	44,460	51,858	68,312	29.57%
Mississippi	29,180	41,040	62,279	61,843	30.57%
Maine	83,340	73,524	71,988	60,803	-9.80%
Delaware	19,260	8,608	38,887	57,954	115.16%
New Hampshire	66,820	58,973	64,345	57,229	-4.56%
Washington DC	34,340	45,912	58,990	49,005	15.09%
South Dakota	39,600	33,576	30,933	34,218	-4.15%
Montana	26,900	25,270	32,166	26,226	0.92%
North Dakota	57,340	34,025	29,824	23,851	-24.35%
Wyoming	19,480	18,972	19,903	23,346	6.53%
West Virginia	25,980	23,580	32,328	22,881	-0.45%
Vermont	21,500	17,524	21,444	21,094	0.75%
State not identified	0	121,137	0	0	NA
Total	14,083,320	18,736,328	24,404,745	28,713,690	26.98%

Note. Source: The U.S. Census/ACS from the IPUMS.

Table 4 indicates rather inconsistent patterns of HL speakers' demographics at the state level, which is corroborated by other types of statistics. The number of HL speakers is one indicator of the presence of adult HL speakers in each state, but it has an inherent bias toward larger states. The true representation of HL speakers can be better represented by the proportion of HL speakers, controlling for the difference in the populations of each state. Table 5 shows such a statistic, the proportions of HL speakers to the state populations.

TABLE 5

The Proportions of HL Speakers to the State Population for each U.S. State in 1980, 1990, 2000,

and 2010

State	1980	1990	2000	2010	Difference between 2010 and 1980	State	1980	1990	2000	2010	Difference between 2010 and 1980
Alabama	1.18%	1.87%	2.38%	2.77%	1.58%	Montana	3.40%	3.17%	3.56%	2.65%	0.75%
Alaska	7.80%	8.07%	8.28%	9.57%	1.76%	Nebraska	3.20%	3.09%	4.40%	4.43%	1.23%
Arizona	11.11%	11.96%	13.16%	12.77%	1.66%	Nevada	6.12%	7.66%	10.86%	11.69%	5.57%
Arkansas	1.16%	1.60%	2.65%	3.29%	2.13%	New Hampshire	7.27%	5.34%	5.21%	4.35%	-2.93%
California	12.10%	16.37%	16.54%	16.72%	4.62%	New Jersey	8.60%	10.13%	11.97%	13.05%	4.45%
Colorado	7.05%	6.94%	8.37%	8.58%	1.53%	New Mexico	22.96%	23.93%	23.00%	22.71%	-0.25%
Connecticut	8.05%	7.83%	8.96%	10.14%	2.09%	New York	10.78%	11.28%	13.08%	13.20%	2.42%
Delaware	3.26%	1.30%	4.97%	6.44%	3.18%	North Carolina	1.54%	2.37%	4.46%	5.39%	3.85%
Washington DC	5.35%	7.53%	10.32%	8.11%	2.75%	North Dakota	8.74%	5.36%	4.64%	3.54%	-5.20%
Florida	7.17%	8.17%	10.09%	11.93%	4.76%	Ohio	2.94%	3.16%	3.48%	3.56%	0.63%
Georgia	1.65%	2.70%	5.35%	6.30%	4.65%	Oklahoma	2.74%	3.40%	4.10%	4.75%	2.01%
Hawaii	16.87%	15.46%	14.19%	12.08%	-4.79%	Oregon	3.28%	4.22%	6.19%	6.61%	3.33%
Idaho	3.29%	3.47%	4.75%	4.95%	1.67%	Pennsylvania	4.10%	4.26%	4.58%	5.23%	1.13%
Illinois	6.27%	7.41%	8.94%	9.70%	3.43%	Rhode Island	10.54%	9.67%	9.47%	8.85%	-1.69%
Indiana	2.41%	2.80%	3.74%	4.21%	1.80%	South Carolina	1.53%	2.19%	3.01%	3.44%	1.90%
Iowa	2.15%	2.34%	3.28%	3.73%	1.58%	South Dakota	5.71%	4.84%	4.10%	4.19%	-1.52%
Kansas	3.20%	3.59%	4.62%	5.25%	2.05%	Tennessee	1.24%	1.74%	2.87%	3.47%	2.23%
Kentucky	1.07%	1.32%	2.35%	2.51%	1.45%	Texas	11.58%	14.20%	15.73%	16.43%	4.86%
Louisiana	7.43%	7.31%	6.32%	5.61%	-1.82%	Utah	4.24%	4.75%	6.87%	7.04%	2.80%
Maine	7.41%	5.99%	5.64%	4.58%	-2.83%	Vermont	4.18%	3.13%	3.52%	3.37%	-0.81%
Maryland	3.68%	5.30%	6.78%	8.08%	4.40%	Virginia	2.85%	4.63%	6.01%	7.35%	4.50%
Massachusetts	7.67%	8.22%	9.57%	10.40%	2.73%	Washington	4.08%	5.19%	7.11%	8.31%	4.23%
Michigan	3.83%	3.92%	4.58%	4.38%	0.55%	West Virginia	1.33%	1.32%	1.79%	1.23%	-0.10%
Minnesota	3.81%	3.37%	4.52%	5.30%	1.49%	Wisconsin	3.60%	3.26%	3.93%	4.13%	0.53%
Mississippi	1.15%	1.60%	2.19%	2.08%	0.93%	Wyoming	4.12%	4.19%	4.03%	4.14%	0.01%
Missouri	1.93%	2.18%	2.92%	3.38%	1.45%	State not identified	NA	NA	NA	NA	NA
						Average of all states	6.21%	7.55%	8.67%	9.28%	3.07%

Note. Source: The U.S. Census/ACT from the IPUMS.

Unsurprisingly, states with large immigrant hubs tend to have relatively high proportions of adult HL speakers. For example, 16.72% of California's state population is comprised of HL speakers; this figure is 16.43% for Texas, 13.20% for New York, 11.93% for Florida, and 9.70% for Illinois. Notably, some states that are not ranked high in raw number of HL speakers have equally large proportions of HL speakers in their population. For example, New Mexico is ranked the 14th largest state in terms of the raw number of HL speakers, but as many as 22.71% of their state population is identified as HL speakers. The same pattern is observed in Connecticut (10.14%), Nevada (11.69%), Hawaii (12.08%), Rhode Island (8.85%), and the District of Columbia (8.11%). A proportion of HL speakers to the state population is critical

information because it has a significant effect on securing each state's public services for HL speakers. The high proportion of adult HL speakers should be interpreted as a need for a larger allocation of public resources for HL speakers in these states, such as bilingual education programs and bilingual access to public services.

The total number of HL speakers is a useful statistic, but it obfuscates the *composition* and *density* of adult HL speakers, two important aspects of HL speakers' demographics, and falls short of demonstrating the complexity of the demographics of adult HL speakers.

Composition of HL Speakers

The raw number of HL speakers does not show the composition of HLs in each region. As the race/ethnicity data suggest, Hispanics account for a significant proportion of total HL speakers, and the 2000–2010 period saw an influx of Hispanic and Asian immigrants. This pattern should be reflected in the composition of HLs in the United States; that is, Spanish should be the largest HL community in the United States, followed by Asian languages, such as Chinese. Using the home language data, we can obtain such linguistically fine-grained estimates of the number of HL speakers in various languages (see Table 6).

TABLE 6
The Numbers of HL Speakers by Language in 1980, 1990, 2000, and 2010

Language	1980	1990	2000	2010	Percentage in 2010	Average Increase rate per decade
Spanish	6,465,280	9,421,958	13,661,063	17,013,399	59.25%	38.42%
French	1,160,560	1,365,599	1,372,867	1,256,193	4.37%	3.23%
Chinese	375,800	776,851	1,005,682	1,233,957	4.30%	52.96%
Hindi and related	163,760	394,952	746,244	1,185,354	4.13%	96.32%
Filipino, Tagalog	354,800	648,741	729,341	789,915	2.75%	34.53%
German	966,920	966,303	893,169	715,471	2.49%	−9.18%
Vietnamese	113,800	330,910	485,837	540,677	1.88%	82.96%
Korean	187,700	429,364	445,707	487,432	1.70%	47.31%
Russian	88,000	154,015	449,141	481,380	1.68%	91.27%
Arabic	146,140	239,654	338,353	439,744	1.53%	45.05%
Dravidian	26,900	65,199	194,481	423,649	1.48%	152.83%
Italian	974,940	796,184	578,891	393,700	1.37%	−25.87%
Portuguese	206,180	235,376	281,333	345,253	1.20%	18.80%
Sub Saharan Africa	35,820	80,853	183,040	335,973	1.17%	111.89%
Polish	602,880	520,920	430,122	310,821	1.08%	−19.59%
Japanese	245,420	295,689	312,001	254,039	0.88%	2.47%
Serbo Croatian, Yugoslavian, Slavonian	92,900	78,208	128,736	182,954	0.64%	30.30%
Greek	232,740	215,638	186,452	159,108	0.55%	−11.85%
Persian, Iranian, Farsi	80,760	147,809	139,201	146,202	0.51%	27.41%

Navajo Navaho	NA	99,298	126,387	139,291	0.49%	18.75%
Thai, Siamese, Lao	56,000	131,887	106,895	108,458	0.38%	39.34%
Amharic, Ethiopian, etc	8,180	26,107	55,837	108,221	0.38%	142.28%
Hebrew, Israeli	60,100	89,757	114,796	106,610	0.37%	23.37%
Indonesian	11,040	29,582	40,959	97,439	0.34%	114.77%
Armenian	64,620	90,430	95,675	95,593	0.33%	15.22%
Micronesian, Polynesian	34,260	41,264	41,411	87,711	0.31%	44.20%
Yiddish, Jewish	149,420	111,973	100,281	82,608	0.29%	-17.71%
Dutch	81,720	75,921	88,776	80,717	0.28%	0.25%
Albanian	7,000	10,394	43,827	77,777	0.27%	149.20%
Ukrainian, Ruthenian, Little Russian	48,400	45,199	64,437	77,290	0.27%	18.63%
Rumanian	20,140	44,701	68,141	76,918	0.27%	62.42%
Tibetan	5,880	34,069	43,552	75,687	0.26%	193.67%
Other East Southeast Asian	8,760	64,685	67,974	67,401	0.23%	214.22%
Turkish	18,460	27,618	46,585	66,476	0.23%	53.66%
Magyar, Hungarian	89,240	72,088	56,916	39,480	0.14%	-23.63%
Lithuanian	43,780	37,496	28,938	34,534	0.12%	-5.95%
Swedish	60,620	49,218	45,134	33,126	0.12%	-17.90%
Czech	90,560	67,121	54,993	32,438	0.11%	-28.32%
Norwegian	78,860	57,130	38,242	27,090	0.09%	-29.93%
Slovak	63,100	66,629	29,751	21,179	0.07%	-26.19%
Other or not reported	185,960	6,993	163,647	18,810	0.07%	685.14%
Finnish	51,540	41,132	28,868	17,999	0.06%	29.22%
Others	110,740	291,879	291,062	445,616	1.55%	72.13%
American Indian, all	213,640	NA	NA	NA	NA	NA
ALL	13,869,680	18,776,794	24,404,745	28,713,690	100.00%	27.67%

By looking at the average increase rates, it becomes obvious that there are two groups of HL speakers in terms of the average growth rate per decade between 1980 and 2010. On one hand, speakers of French (3.23%), German (-9.18%), Italian (-25.87%), Portuguese (18.80%), Polish (-19.59%), Japanese (2.47%), and Greek (-11.85%) are undergoing declines in the number of HL speakers or much smaller growth than the nation's average growth rate of 27.67%. The other group is HL speakers of languages spoken by recent immigrants such as Spanish (38.42%), Chinese (52.96%), Hindi (96.32%), Filipino/Tagalog (34.53%), Vietnamese (82.96%), Korean (47.31%), Russian (91.27%), Arabic (45.05%), and Dravidian (152.83%). This growth trajectory from 1980 to 2010 predicts that some of the commonly spoken HLs in the United States (e.g., French, German, Italian, Portuguese, and Japanese) may diminish in the next few decades. In fact, according to a recent data by Fee et al. (2014), Chinese has already replaced French and become the second most commonly spoken HL, and Vietnamese has replaced German as the sixth most commonly spoken HL.²

Additionally, the ranking of the HLs in Table 6 might not match our intuitive estimation. For example, Filipino/Tagalog, the fifth largest HL group in the United States, is not very

common in New York. Italian, on the contrary, is one of the typical HLs in New York, but it is barely found in West Coast states, such as California. Navajo, the 30th most common HL in the United States, is one of the most typical HLs in Arizona and New Mexico, but it is nonexistent outside of these two states. Obviously, there are many regional and geographic variations in the distribution of HLs in the United States.

To illustrate these regional differences, the fifteen most common HLs in California, Texas, New York, Florida, and New Mexico are plotted in Figure 3. In California, Spanish (63.41%), Chinese (6.51%), Filipino/Tagalog (65.56%), Vietnamese (2.80%), and Korean (2.52%) account for higher proportions than the national averages. On the contrary, there are fewer proportions of French, German, Italian, Arabic, Hindi, Russian, Portuguese, and Polish in California.

The composition of HLs in New York is rather stark. One noticeable difference is that the proportion of Spanish speakers in New York (48.00%) is much lower than the national average. Additionally, in New York, French (5.87%), Italian (3.83%), Russian (4.33%), and Polish (1.91%) have higher proportions than the national averages. These languages are less common in California, exhibiting complementary patterns of HLs between California and New York. Another interesting pattern observed in the comparisons of HLs in California and New York is the proportion of Spanish HL speakers. Spanish is the most frequently spoken HL in both states, but the proportions are different. In California, Spanish HL speakers account for 63.41% of the total HL speakers, whereas they account for 48.00% in New York. In New York, the HLs other than Spanish collectively represent more than half of the HL speakers (nearly 50%) in the state.

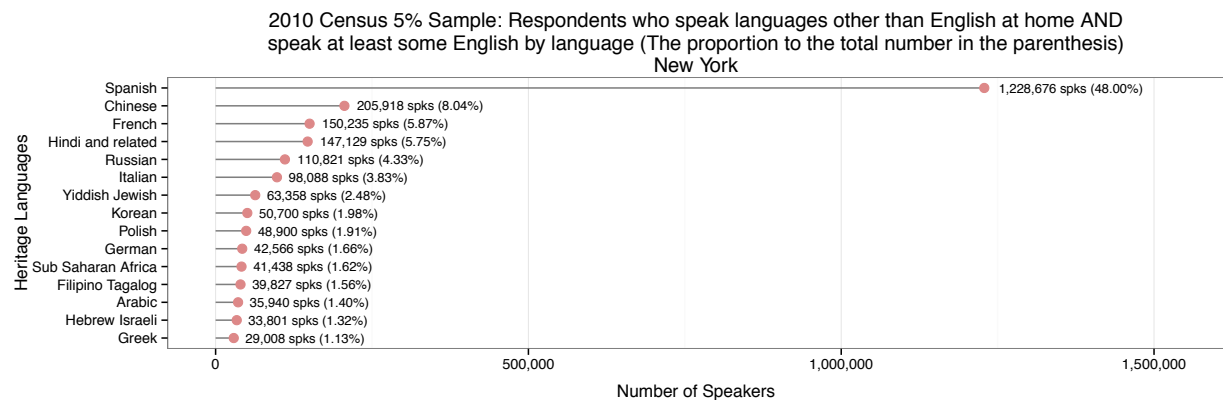
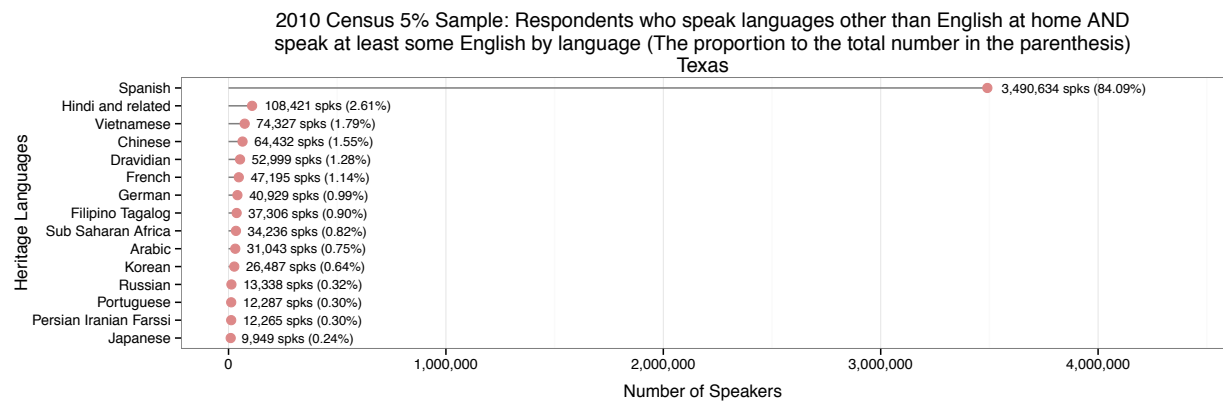
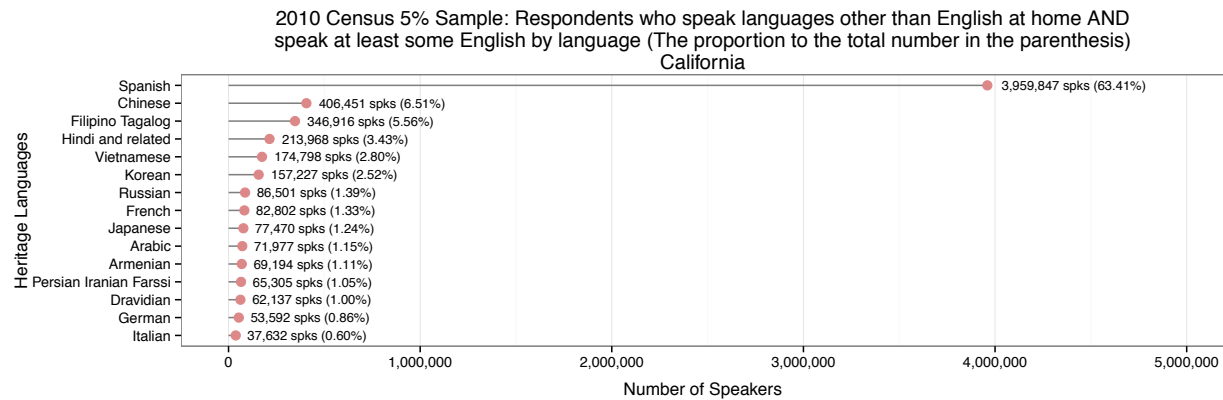
Texas contrasts greatly with New York in its distribution of HLs because Spanish HL speakers account for as much as 84.09% of the total HL speakers in Texas. In Texas, Chinese, which is the second most commonly spoken HL in California and New York, has only a marginal representation (1.55%) and is outnumbered by Hindi and Vietnamese.

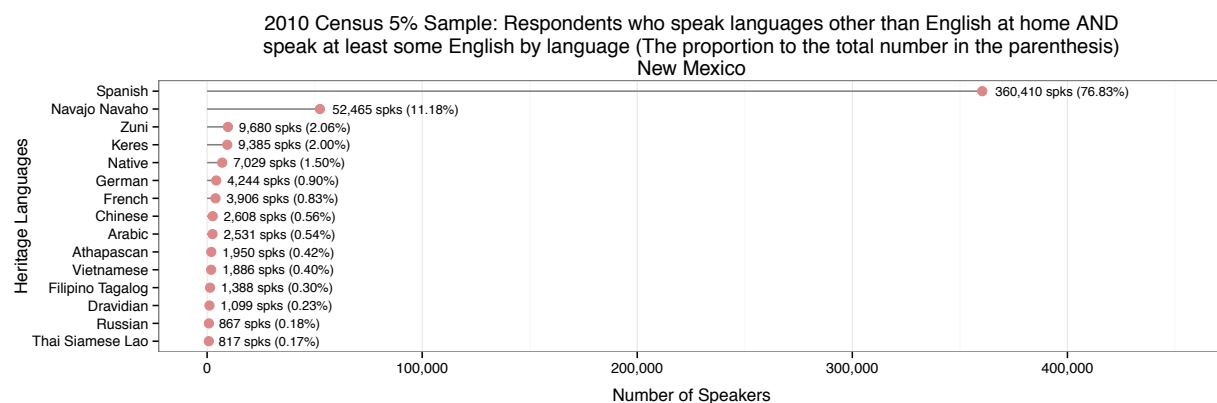
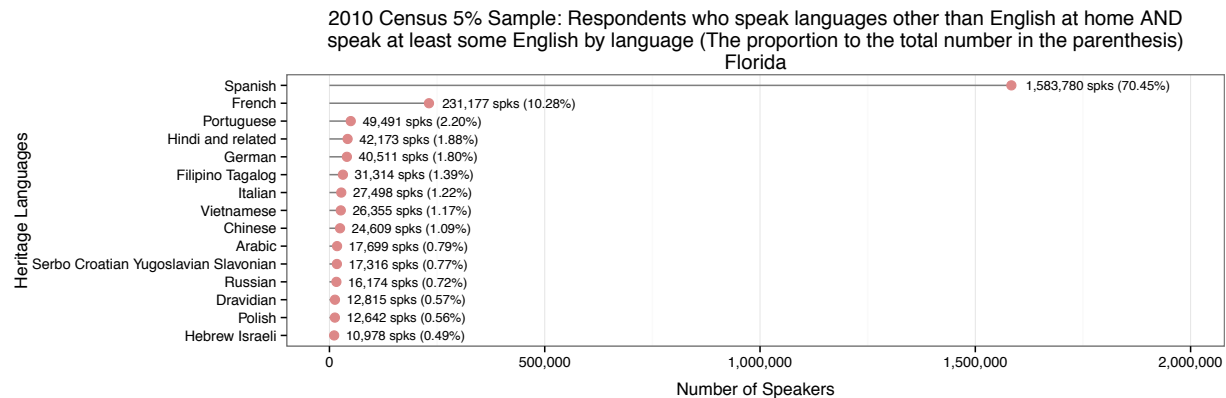
In fact, the comparison of these states makes it clear that the only shared pattern is that Spanish is the most commonly spoken HL. Other than this single fact, almost everything else, such as the proportions of HL speakers and the rankings of commonly spoken HLs, radically differs from state to state. For example, in Florida, Spanish accounts for 70.45% of the total HL speakers, followed by French (10.28%) and Portuguese (2.20%). In New Mexico, Spanish accounts for 76.83%, followed by Navajo (11.18%) and Zuni (2.06%). As discussed above, in New York, Spanish accounts for much smaller proportion (only 48%) than those of Florida and New Mexico, followed by Chinese (8.04%), French (5.87%), and Hindi (5.75%).

In addition, if we look at all the U.S. states, we can find quite a few exceptions where Spanish is not the most common HL, such as Alaska (Aleut Eskimo is the most common HL), Hawaii (Filipino/Tagalog and Japanese), Louisiana (French), Maine (French), New Hampshire (French), North Dakota (German), South Dakota (Siouan languages), and Vermont (French). Thus, in sum, this survey of the composition of HLs reveals no consistent pattern at the state level.

FIGURE 3

The Fifteen Most Commonly Spoken HLs in California, Texas, New York, Florida, and New Mexico





Density of HL Speakers

Another aspect of the demographics of HL speakers is their density in a specific region. The raw number of HL speakers does not show how dense the population of HL speakers is in each community. Information about density is particularly crucial when we determine the distribution of resources for HL speakers because the availability of resources, such as community language schools, must be located within an accessible, commutable distance from where HL speakers reside. For example, there are approximately 2.6 times more HL speakers in California than in New York, but in terms of geographic area, California (155,959 square miles) is 3.3 times larger than New York (47,213 square miles). Therefore, the raw numbers of HL speakers may not directly result in a higher density of HL speakers in California than in New York.

In this study, the Gini index is employed to investigate the density of HL speakers. The Gini index is a statistic of dispersion and shows the degree of distribution in a range from zero to one. A Gini index of zero indicates an equal spread (i.e., HL speakers are equally distributed across all ConsPUMAs/counties), and a Gini index of one indicates the most skewed spread (i.e., HL speakers are centered in one single ConsPUMA/county). Table 7 shows the Gini indexes for all HLs.

TABLE 7
GINI Index of HL Speakers for Each HL

Language	1980	2000	2010	Change 1980– 2010	Language	1980	2000	2010	Change 1980– 2010
Spanish	0.828	0.769	0.764	−0.063	Rumanian	0.902	0.843	0.840	−0.062
French	0.774	0.728	0.766	−0.007	Tibetan	0.918	0.940	0.948	0.031
Chinese	0.853	0.841	0.821	−0.032	Hamitic	0.935	0.952	0.950	0.015
Hindi and related	0.778	0.783	0.786	0.008	Other East Southeast Asian	0.911	0.892	0.913	0.003
Filipino Tagalog	0.875	0.854	0.847	−0.029	Turkish	0.839	0.813	0.839	0.000
German	0.615	0.569	0.621	0.006	Native	NA	NA	0.929	NA
Vietnamese	0.797	0.835	0.845	0.048	Other Balto Slavic	0.900	0.843	0.902	0.002
Korean	0.798	0.825	0.854	0.055	Magyar Hungarian	0.810	0.754	0.852	0.042
Russian	0.831	0.833	0.808	−0.023	Lithuanian	0.853	0.825	0.897	0.043
Arabic	0.767	0.749	0.773	0.006	Near East Arabic dialect	0.914	0.959	0.972	0.058
Dravidian	0.811	0.798	0.820	0.009	Swedish	0.768	0.754	0.867	0.099
Italian	0.804	0.752	0.765	−0.038	Czech	0.835	0.800	0.877	0.042
Portuguese	0.904	0.851	0.855	−0.048	Other Malayan	0.902	0.903	0.940	0.038
Sub Saharan Africa	0.809	0.815	0.816	0.007	Athapaskan	NA	0.986	0.953	NA
Polish	0.815	0.810	0.854	0.039	Burmese Lisu Lolo	0.963	0.937	0.927	−0.036
Japanese	0.859	0.795	0.822	−0.037	Norwegian	0.868	0.794	0.895	0.027
Serbo Croatian Yugoslavian Slavonian	0.825	0.767	0.837	0.011	Aleut Eskimo	NA	0.997	0.997	NA
Greek	0.791	0.773	0.803	0.012	Slovak	0.885	0.822	0.881	−0.004
Persian Iranian Farsi	0.786	0.878	0.900	0.113	Other or not reported	0.595	0.767	0.893	0.298
Navajo Navaho	NA	0.985	0.990	NA	Other Persian dialects	0.933	0.941	0.950	0.017
Thai Siamese Lao	0.771	0.769	0.839	0.068	Finnish	0.874	0.841	0.916	0.042
Amharic Ethiopian etc	0.926	0.903	0.912	−0.013	Hawaiian	0.957	0.958	0.980	0.024
Hebrew Israeli	0.882	0.855	0.895	0.013	Siouan languages	NA	0.940	0.985	NA
Indonesian	0.872	0.834	0.867	−0.006	Danish	0.775	0.787	0.902	0.126
Armenian	0.931	0.955	0.965	0.034	Celtic	0.770	0.817	0.917	0.147
Micronesian Polynesian	0.919	0.895	0.925	0.006	Zuni	NA	NA	0.998	NA
Yiddish Jewish	0.920	0.912	0.972	0.052	Keres	NA	0.993	0.998	NA
Dutch	0.736	0.676	0.799	0.064	African NS	0.901	0.909	0.967	0.066
Albanian	0.932	0.899	0.912	−0.020	Iroquoian	NA	0.914	0.969	NA
Ukrainian Ruthenian Little Russian	0.825	0.808	0.881	0.056	American Indian NS	NA	NA	0.960	NA
Average						0.844	0.846	0.886	0.026

The average Gini index was 0.886 in 2010, indicating that HL speakers usually reside in

specific areas of each state. This is not surprising because it is a typical pattern of demographics that populations cluster around large cities. The Gini index also shows the different degrees of distribution across the HL groups. The race/ethnicity data show that Hispanic communities are more spread out across counties than the Chinese, Filipino/Tagalog, Korean, Portuguese, and Japanese communities are. The Gini index for Spanish HL speakers was 0.764 in 2010, the second to the lowest after German HL speakers (0.621). In comparison, HL speakers of Chinese (0.821), Filipino/Tagalog (0.847), Vietnamese (0.845), Korean (0.854), Dravidian (0.820), Portuguese (0.855), and Japanese (0.822) are more densely concentrated than are Spanish and German communities. The transition of the density of HL communities from 1980 to 2010 shows that the HL community as a whole has moved away from the high-density immigrant enclaves and spread towards the suburbs and outside of the central cities. This tendency is most obvious with the Spanish community, whose Gini index reduced by 0.063 from 0.828 in 1980 to 0.764 in 2010.

SUMMARY OF DISCUSSION

As presented at the beginning of this study, the survey of the U.S. Census and ACS data from 1980 to 2010 reveals the following demographic patterns of HL speakers in the United States:

- (1) The number of HL speakers grew at a considerably faster rate (26.98% per decade from 1980–2010) than the average growth of the U.S. population (10.88% per decade from 1980–2010). However, the growth rates radically differ from state to state. In some states, there was even a decline in the number of adult HL speakers.
- (2) Spanish and Chinese remain the two most common groups of HLs in the United States, and their prevalence has grown rapidly over the last 30 years. In addition, although their numbers are still small in relation to the absolute number of speakers, new HLs such as Arabic, Hindi, Dravidian, Vietnamese, Russian, Amharic/Ethiopian, and Tibetan, also experienced substantial growth. At the same time, languages such as French, German, Italian, Greek, Yiddish, and Dutch are experiencing rapid declines.
- (3) Unsurprisingly, states that are typically considered immigration hubs (e.g., California, Texas, New York, Florida, and New Mexico) have large numbers of HL speakers. However, in terms of the proportion of HL speakers in the state's population, some other states (e.g., Arizona, Hawaii, Massachusetts, Nevada, and New Jersey) have proportions of HL speakers as large as those in states with high numbers of immigrants.
- (4) The majority of HL speakers are speakers of Spanish, but on the state level, there are quite a few exceptions, such as Alaska (Aleut Eskimo is the most common HL), Hawaii (Filipino/Tagalog and Japanese), Louisiana (French), Maine (French), New Hampshire (French), North Dakota (German), South Dakota (Siouan languages), and Vermont (French).
- (5) Over the last 30 years, the concentration of HL speakers has moved from metropolitan centers to suburbs or non-metropolitan areas. However, some of the HLs continue to have specific regional concentrations.

At the national level, we observe the transition from the languages former immigrants groups (e.g., French, German, Italian, Greek, Yiddish, and Dutch) to the languages of recent

immigrants, particularly Spanish and Chinese. However, the data also show a sign of incoming trends in the next few decades. Languages such as Arabic, Hindi, Dravidian, Vietnamese, Russian, Amharic/Ethiopian, and Tibetan are growing even faster than Spanish and Chinese, forecasting that, over the next few decades, that the numbers of speakers of these languages may grow to be as large as that of Spanish or Chinese languages.

At the state or county level, the data reveal a great deal of diversity in the demographics of adult HL speakers. Each state has a unique composition of HL speakers, and the national trends do not apply to all the states and counties. For example, the second most commonly spoken HL at the national level is Chinese, but as illustrated by the compositions of HLs at the state level, this fact applies to only five out of the fifty total states and one special district (9.8%). It is a widely acknowledged fact that Spanish is by far the most common HL in the United States, but, in eight states (15.6%), another language is spoken more frequently than Spanish. Even among the states where Spanish is spoken most commonly, the proportions of Spanish HL speakers vary considerably from 38.38% (Pennsylvania) to 76.83% (New Mexico).

It is important that research on HL keep note of such regional differences. In particular, studies regarding the attrition or maintenance of HLs will lose their generalizability if they do not take into consideration the influence of the regional demographics of HL speakers where their data were collected. Using multivariate modeling, Alba et al. (2002) have demonstrated that the HL community has the most significant influence on the maintenance of HLs (other than whether their parents are endogamously married). Thus, findings on the attrition or maintenance of HL in one place (for instance, Philadelphia, where a relatively smaller Spanish community exists) may not be applicable to another place (e.g., New Mexico, where nearly three-fourths of HL speakers speak Spanish). Obviously, studies on HL in general should acknowledge such geographic limitations and should carefully select their data collection sites so that they are counterbalanced in terms of the demographics of HL speakers.

The density of HL speakers has shown the outward migration of HL speakers in the last three decades. Ehrenhalt (2013) uses the term “demographic inversion” (p. 3) to describe this trend of new Hispanic (and black) immigrants such that first- and 1.5-generation immigrants directly settle in suburbs rather than in central cities. Central cities, which used to offer manufacturing workplaces and affordable housing for the poor and newly arrived immigrants, have undergone gentrification and are now attracting wealthy and middle-class residents. As far as the HL speakers of Spanish are concerned, the outward migration of the affluent from cities to suburbs has come to an end, and many of them have moved to suburbs much more quickly than before. On the contrary, some groups of HL speakers, such as Chinese, Filipino/Tagalog, Korean, Vietnamese, Korean, Dravidian, and Portuguese, still maintain the traditional outward migration model and reside in the ethnic enclaves with a high density of HL speakers.

This difference between Spanish and other HL speakers, particularly speakers of Chinese languages, (also see Pew Research Center, 2013) is an interesting parallel with the findings that Asian (Chinese) HL speakers have a faster attrition rate than Spanish HL speakers (Lopez, 1996; Alba et al., 2002). Typically, these different speeds of language attrition are attributed to the typological distance between the dominant language (i.e., English) and the HL (i.e., Spanish is typologically much closer to English than Chinese), but this study suggests an alternative interpretation of the higher maintenance rate of Spanish as an HL; that is, Spanish HL speakers’ unique emigration pattern from the metropolitan areas in the recent history. It is an empirical question whether the higher maintenance of HL is caused by the influence of the demographics

of HL communities or the linguistic similarities between the mainstream language and the HL.

It goes without saying that maintenance of HLs involves many other factors, including HL speakers' language ideologies, degree of cultural assimilation, immigration age, religious participation in the HL, continuing input in the HL through ethnic media or trips to the country of origin, motivation, social prestige of the HL, market use/value of the HL, availability of HL courses/bilingual programs, and isolation or existence of ethnic enclaves, just to name a few. In recent studies, Lo Bianco (2014) and Lo Bianco and Payton (2013) argue that creating an environment to increase vitality and desire to use the HL is critical for maintaining the HL.

It is very likely that there is no single deterministic factor that can predict the likelihood of HL speakers' maintenance of their HL, since many of these factors overlap with each other within a complex network of facilitative and hindering relationships. This study has demonstrated that demographic factors of HL speakers, particularly their absolute numbers and density, exhibit variation by both region and language. As future research, if one can gather data for non-demographic variables mentioned above for various language groups in different geographic locations, it is possible to develop and examine statistical models of these complex relationships of variables in HL maintenance.

CONCLUSION

Recently, HL speakers have received considerable attention in multi-disciplinary research in language education, language acquisition, and language policy. This article has presented the demographics of HL speakers, which are crucial in making decisions about research projects, allocating resources for HL speakers, and establishing curricula at an educational institution. I have argued that in addition to the raw number of HL speakers, their composition and density need to be considered in understanding the demographics of HL speakers in the United States.

These findings of HL speakers' demographics are also useful for future research on HLs. This study suggests the possibility that the different attrition rates of HLs between Spanish and Asian languages may be attributed to the different demographic patterns between these two language groups rather than to the HLs' linguistic typological distance from English.

NOTES

¹ The 1.5 generation in Rumbaut et al. (2005, 2009) is defined as immigrants who arrive in the United States before the age of 12. Although the majority of studies define the 1.5-generation immigrants as those who arrived in the United States during school age, but the exact cut-off age varies from study to study.

² These figures were collected from the ACS data in 2005 and 2007. Unlike this study, in which the target population is adult bilingual HL speakers, the rankings by Fee et al. (2014) are based on the numbers of all HL speakers (including young children and first-generation HL speakers). This different approach to defining HL speakers likely caused these minor differences in the rankings of HL speakers as of 2005–2007.

REFERENCES

- Alba, R., Logan, J., & Stults, B. (2002). Only English by the third generation? Loss and preservation of the mother tongue among the grandchildren of contemporary immigrants. *Demography*, 39, 467–484.
- Au, T-K-F., Knightly, L. M., Jun, S-A., & Oh, J-S. (2002). Overhearing a language during childhood. *Psychological Science*, 13, 238–243.
- Brecht, R. D. & Ingold, C. W. (2002). Tapping a national resource: Heritage languages in the United States. Center for Applied Linguistics: Washington D.C.
- Carreira, M., Jensen, L., Kagan, O., & Giangrande, M. (2009). The Heritage Language Learner Survey: Report on the Preliminary Results. Accessed 2 March 2015 at <http://www.international.ucla.edu/media/files/paper.pdf>
- Carreira, M., & Kagan, O. (2011). The results of the National Heritage Language Survey: Implications for teaching, curriculum design, and professional development. *Foreign Language Annals*, 44, 40–64.
- Ehrenhalt, A. (2013). *The great inversion and the future of the American city*. New York: Vintage International.
- Fee, M., Rhodes, N. C., & Wiley, T. G. (2014). Demographic realities, challenges, and opportunities. (1). In T. G. Wiley, J. K. Peyton, D. Christian, S. C. K. Moore, & N. Liu (Eds.), *Handbook of heritage, community, and Native American languages in the United States: Research, policy, and educational practice* (pp. 6–18). New York: Routledge.
- Fishman, J. A. (2001). 300-plus years of heritage language education in the United States. In J. K. Peyton, D. A. Ranard, & S. McGinnis (Eds.), *Heritage languages in America: Preserving a national resource* (pp. 81–98). Washington, DC: Center for Applied Linguistics.
- Gaquin, D. A. & White–Dunn, G. (2014). *The who, what, and where of America: Understanding American Community Survey* (4). Lanham, MD: Bernan Press.
- Hornberger, N. H., & Wang, S. C. (2008). Who are our heritage language learners? Identity and biliteracy in heritage language education in the United States. In D. Brinton, O. Kagan, & S. Bauckus (Eds.), *Heritage language education. A new field emerging* (pp. 3–35). New York: Routledge.
- Kagan, O., & Dillon, C. (2003). Heritage speakers' potential for high-level proficiency. In H. Byrnes & H. H. Maxim (Eds.), *Advanced foreign language learning: A challenge to college programs* (pp. 99–112). Boston, MA: Heinle/Thomson.
- Lo Bianco, J. & Peyton, J. K. (2013). Vitality of Heritage Languages in the United States: The Role of Capacity, Opportunity, and Desire. *Heritage Language Journal*, 10(3).
- Lo Bianco, J. (2014). Language vitality and heritage languages. In T. G. Wiley, J. K. Peyton, D. Christian, S. C. K. Moore, & N. Liu (Eds.), *Handbook of Heritage, Community, and Native American Languages in the United States: Research, Policy, and Educational Practice*. (pp. 54–65). New York, NY: Routledge.
- Lopez, D. E. (1996). Language: Diversity and assimilation. In R. Waldinger & M. Bozorgmehr

- (Eds.), *Ethnic Los Angeles* (pp. 139–159). New York: Russell Sage Foundation Publications.
- Montrul, S. (2002). Incomplete acquisition and attrition of Spanish tense/aspect distinctions in adult bilinguals. *Bilingual Language and Cognition*, 5, 39–68.
- Montrul, S. (2009). Knowledge of tense—aspect and mood in Spanish heritage speakers. *The International Journal of Bilingualism*, 13, 239–269.
- Mori, Y., & Calder, T. M. (2013). Bilingual vocabulary knowledge and arrival age among Japanese heritage language students at Hoshuukoo. *Foreign Language Annals*, 46, 290–310.
- Oh, J. S., Knightly, L., & Au, T.–K.–F. (2003). Holding on to childhood language memory. *Cognition*, 86, B53–B64.
- Pan, S. (2002). Chinese in New York. In O. García & J. Fishman (Eds.), *The multilingual apple: Languages in New York City* (pp. 231–255). New York: Mouton de Gruyter.
- Pew Research Center, 2013. “The Rise of Asian Americans”. Pew Social Trends: <http://www.pewsocialtrends.org/2013/04/04/asian-groups-in-the-u-s/>
- Polinsky, M., & Kagan, O. (2007). Heritage languages: In the ‘wild’ and in the classroom. *Language and Linguistics Compass*, 1, 368–395.
- Polinsky, M. (2011). Reanalysis in adult heritage language: A case for attrition. *Studies in Second Language Acquisition*, 33, 305–328.
- Potowski, K. (2010). *Language diversity in the USA*. Cambridge, MA: Cambridge University Press.
- Ruggles, S., & Sobek, M. (1997). Integrated public use microdata series: Version 2.0. Minneapolis, MN: Historical Census Projects, University of Minnesota.
- Rumbaut, R. G., Massey, D. S., & Bean, F. D. (2005). Linguistic life expectancies: Immigrant language retention in Southern California. *Population and Development Review*, 32, 447–460.
- Rumbaut, R. G. (2009). A language graveyard? The evolution of language competencies, Preferences and use among young adult children of immigrants. In T. G. Wiley, J. S. Lee, & R. Rumberger (Eds.), *The education of language minority immigrants in the United States* (pp. 35–71). Bristol, UK: Multilingual Matters.
- Ryan, C. (2013). Language use in the United States: 2011. US Census Bureau.
- Valdés, G. (2001). Heritage language students: Profiles and possibilities. In J. K. Peyton, D. A. Ranard, & S. McGinnis (Eds.), *Heritage languages in America: Preserving a national resource* (pp. 37–77). Washington, DC: Center for Applied Linguistics.
- Wiley, T. G., Peyton, J. K., Christian, D., Moore, S. C. K., & Liu, N. (Eds.) (2014). *Handbook of heritage, community, and Native American languages in the United States: Research, policy, and educational practice*. New York: Routledge.
- Zentella, A. C. (1997). *Growing up bilingual: Puerto Rican children in New York City*. Oxford: Basil Blackwell.

Zentella, A. C. (2002). Spanish in New York. In O. García & J. Fishman (Eds.), *The multilingual Apple: Languages in New York City* (pp. 167–201). New York: Mouton de Gruyter.